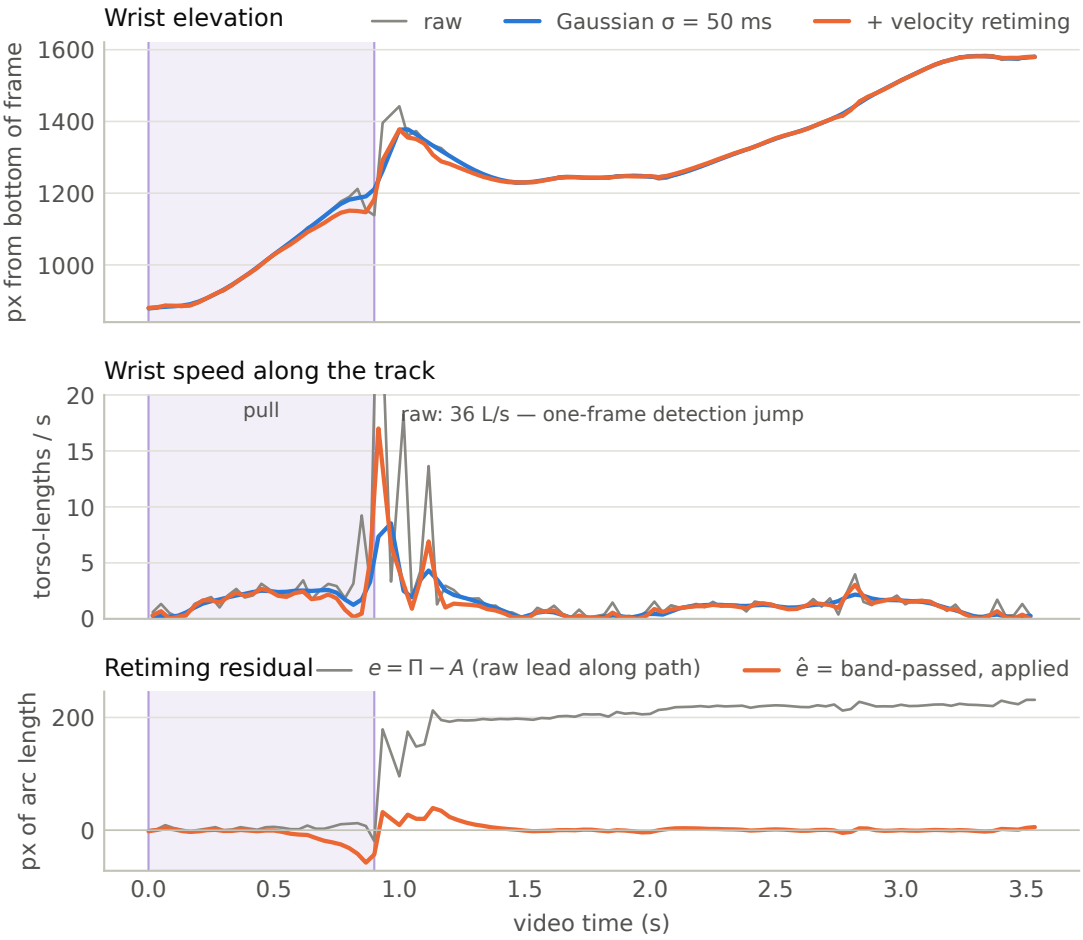

Wrist elevation
raw
Gaussian σ = 50 ms
+ velocity retiming
px from bottom of frame
1600
1400
1200
1000
Wrist speed along the track
pull
raw: 36 L/s — one-frame detection jump
torso-lengths / s
20
15
10
5
0
Retiming residual
e = Π − A (raw lead along path)
ê = band-passed, applied
px of arc length
200
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
video time (s)